

Fine-Tuning LLM Models for Bug Detection in Quantum Software

We fine-tune open-source LLM Models on labeled Qiskit bug data to detect and classify quantum software bugs, benchmarking against prompt-only and RAG baselines.

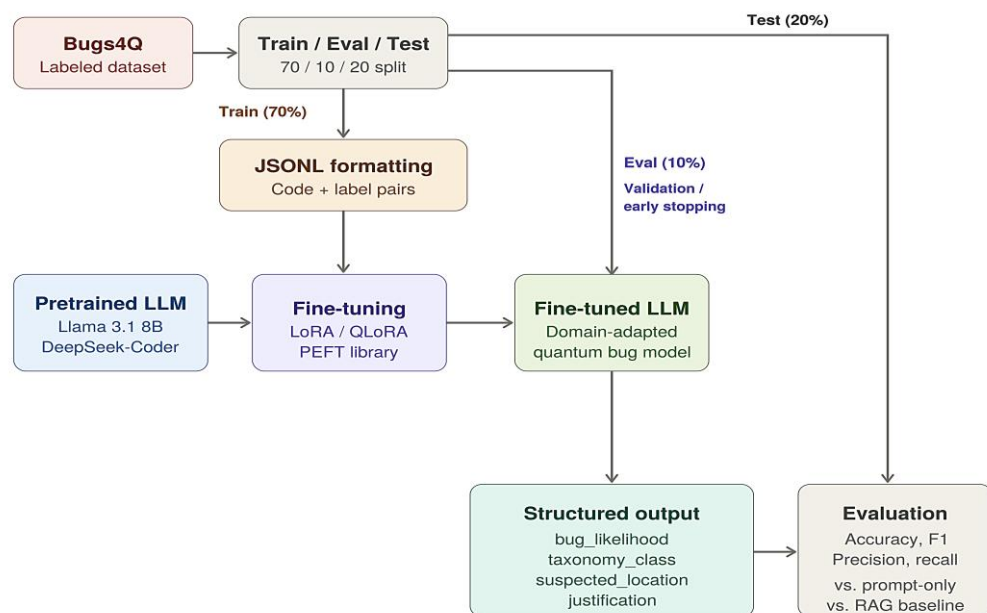
Background & Motivation

- Quantum software contains domain-specific bugs not handled well by classical tools.
- Fine-tuning for quantum bug classification remains underexplored.
- Existing work explores only static approach, prompt-only and RAG baselines.
- Fine-tuning embeds bug patterns into model weights, offering faster and more specialized detection than retrieval-based methods.

Data and Resources

- Bugs4Q** (public, GitHub): real Qiskit bugs with ground-truth labels for evaluation.
- Bugs-QCP** (Zenodo): taxonomy corpus for RAG knowledge-base enrichment only.
- APIs**: LoRA for fine-tuning; OpenAI API for baselines.
- Compute**: single GPU via Colab or university HPC.

Possible Technical Approach



Impact

- First fine-tuning benchmark for Qiskit bug classification.
- Helps build specialized debugging tools for quantum software.
- Quantum bugs in security-critical applications (e.g., QKD) can cause silent failures that compromise cryptographic guarantees.
- Risks**: overfitting (mitigate with LoRA + early stopping), label leakage (strict train/eval splits).

References

- Zhao et al. (2021). Bugs4Q. ASE.
- Campos & Souto (2021). Q Bugs. arXiv.
- Hu et al. (2022). LoRA. ICLR.